

Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **6738**

AVI	2025-26	Session:	Odd MSE October 2025	Date:	08-10-2025
Sem:	1	Slot:	10:00am - 11:00am	C. Code:	CS413
Exam Type:	Written	Block:	202	Seat No:	2022600016
Course Section:	1				



Name of the candidate. Aashay Jaethar

Candidate's UID number:

2	0	2	2	6	0	0	0	1	6
---	---	---	---	---	---	---	---	---	---

Examination class room number:

2	0	2
---	---	---

Invigilator's signature with date:

[Signature] 08/10/2025

(To be filled in by the candidate)

EXAMINATION: BE B.TECH MSE
(For example FY MCA/FY BTECH)

SEMESTER :

PROGRAM: A.I.M.L

I / II / III / IV / V / VI / VII / VIII ✓

DAY AND DATE: WEDNESDAY 8/10/25

COURSE NAME: Deep Learning

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

Space
For
MarksQuestion
No.

START WRITING HERE
(Begin answer for each question on a new page)

1a $f(w) = \frac{1}{2} \|x_w - y\|^2$

$$\vec{w} \leftarrow \vec{w} + \Delta \vec{w}$$

$$\Delta \vec{w} = -n \nabla f(w)$$

$$\therefore \Delta w_i = -n \frac{df}{dw_i}$$

$$= -n \frac{d(\frac{1}{2} \|x_w - y\|^2)}{dw_i}$$

$$= -n \frac{d(\frac{1}{2} \times 2 \times (x_w - y) \times \frac{d(x_w - y)}{dw_i})}{dw_i}$$

$$= -n x_i (x_w - y) \times \frac{d(x_w - x_{wi})}{dw_i}$$

$$= -n (x_w - y) (-x_{id})$$

$$\Delta w_i = n (x_w - y) (x_{id})$$

$$\therefore w_i = w_i + n (x_w - y) (x_{id})$$

Sr. No. 6738



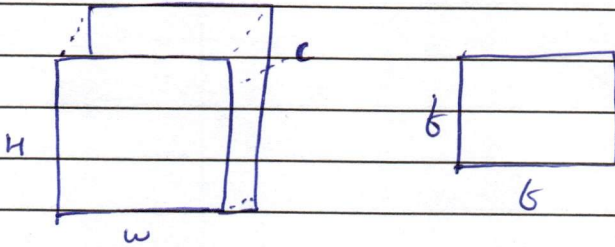
4

[illegible]

Space
For
MarksQuestion
No.

START WRITING HERE
(Begin answer for each question on a new page)

2a



For 2D convolution,

Each filter of $F \times F$ will have 'C' copies and it will perform multiplication at once.

At one instant, no of multiplication = $F \times F \times C$

for K filters = $K \times F \times F \times C$

No. of times these multiplications will happen = $(H - F + 1) \times (W - F + 1)$

$\therefore$ Computational complexity = $(H - F + 1)(W - F + 1) \times K \times F \times F \times C$
= $(H - F + 1)(W - F + 1) K F^2 C$.

This will lead to a dimension of $(H - F + 1)(W - F + 1)$ as output matrix.

In case of depthwise separable convolution, the computational complexity will be $(H - F + 1)(W - F + 1) K F^2 C$ calculated similar to 2D convolution but the final output will have C matrices of $(H - F + 1)(W - F + 1)$ dimension leading to total dimension of $(H - F + 1)(W - F + 1) C$.

Sr. No. 6738



6

[illegible]

Space
For
MarksQuestion
No.

START WRITING HERE
(Begin answer for each question on a new page)

2b A linear undercomplete autoencoder has fewer neurons in the hidden layer as compared to input layer.

This forces the network to learn efficiently compressed representation of the data. It enhances the meaningful understanding of the data.

This leads to reduction of dimensionality. When the autoencoder is trained with MSE, it leads to accurate weight updation leading to in-depth understanding of the data in a very reduced dimensional space.

Thus, the linearly undercomplete autoencoder trained with MSE learns a subspace equivalent to PCA.

Overcomplete autoencoder has more neurons in the latent (hidden) layer as compared to input layer. This ~~leads~~ allows a model to learn or copy the input to output instead of understanding the meaningful representation of data.

This network requires regularization to avoid overfitting and understand the true internal logic and meaning of data.

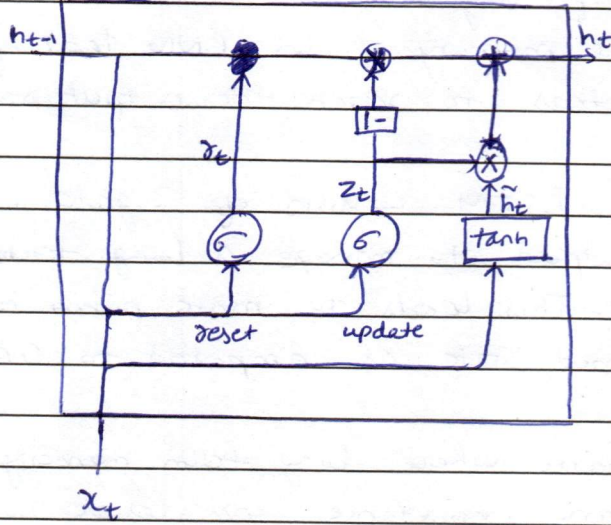
Without regularization, the overcomplete autoencoder thus tends to learn the identity function.

Sr. No. 6738

[illegible]

START WRITING HERE
(Begin answer for each question on a new page)

3b



Equations from the diagram.

$$z_t = \sigma(W_{xz} x_t + W_{hz} h_{t-1} + b_z) \quad -1$$

$$r_t = \sigma(W_{xr} x_t + W_{hr} h_{t-1} + b_r) \quad -2$$

$$\tilde{h}_t = \tanh(W_{xh} x_t + W_{hh} (r_t \times h_{t-1}) + b_h) \quad -3$$

$$h_t = (1 - z_t) h_{t-1} + z_t \tilde{h}_t \quad -4$$

from 4

$$h_t = (1 - z_t) h_{t-1} + z_t \tilde{h}_t$$

$$h_t = h_{t-1} - z_t (h_{t-1} - \tilde{h}_t)$$

$$(h_{t-1}) - h_t = z_t (h_{t-1} - \tilde{h}_t)$$

$$\therefore z_t = \frac{h_{t-1} - h_t}{h_{t-1} - \tilde{h}_t}$$

$$\therefore z_t = \frac{h_{t-1} - h_t}{h_{t-1} - \tilde{h}_t}$$

$$\therefore z_t = \frac{h_{t-1} - h_t}{h_{t-1} - \tilde{h}_t}$$

$\therefore$ GRU's update rule can be interpreted as a convex combination of the previous hidden state (h_{t-1}) & candidate hidden state ($\tilde{h}_t$).

[illegible]



Sr. No. 6738

[illegible]

Sr. No. 6738



Space
For
Marks

Question No.	Answer
--------------	--------

START WRITING HERE
(Begin answer for each question on a new page)



Sr. No. 6738

[illegible]

Sr. No. 6738

[illegible]

[illegible]

